Fiscal Year:	FY 2005	**Task Last Updated:**	FY 03/15/2005
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Project Title:	Granular flow around immersed objects		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Granular flows		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2001 Fluid Physics 01-OBPR-02: Research and Flight Experiment Opportunities
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No. of PhD Candidates:	1	**No. of Master' Degrees:**	1
No. of Master's Candidates:	3	**No. of Bachelor's Degrees:**	2
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA GRC
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	Many granular flows involve the interaction between the granular material and an immersed obstacle. Examples from industry include flow around hopper and bin inserts and structural members, agitating blades in mixers and granulators, heat transfer surfaces in fluidized beds, and Earth-moving surfaces in agricultural and construction applications. Despite being such a common geometry, there remains a lack of fundamental understanding on how to properly model this flow situation, especially in reduced gravity environments. In addition to having many applications, the study of this classic geometry may also lead to further developments and understanding of the governing dynamics of granular flows. This research project involves experiments, numerical computations, and theoretical analyses that investigate the drag forces acting on, and flow fields around, objects immersed in a wide range of granular flow conditions. 1. Additional investigations concerning the transition from dense to dilute granular flow remain to be completed. In particular,		

Task Description:	simulations will be performed using periodic boundaries of a specified width (acknowledging the fact that the measured drag force will be a function of the system size). The force acting on the cylinder will be used as an indicator of transition. 2. The vertical chute, dense granular flow experimental apparatus at Purdue University will be used to investigate the drag force on objects immersed in an accelerating, dense granular flow. Key to the operation of the device is the active control of the exit area so that the flow velocity through the test section will increase proportionally with time giving constant flow acceleration. The relation between the exit opening and the flow velocity is determined using the Beverloo correlation and the apparatus will be controlled using a LabView VI and a PC with an analog output DAQ card. 3. Additional simulations will be performed to investigate granular shock segregation. Rather than use the geometry of a cylinder in a dilute granular flow, a sharp corner geometry will be used so that an oblique shock will be produced. 1. Knowing when a granular flow will transition from dense to dilute flow is critical for predicting the forces acting on objects immersed in a granular flow. The models used to predict the forces depends strongly on the flow regime. 2. Knowing the added mass of an object in an accelerating dense granular flow is critical for predicting unsteady forces. In particular, unsteady dense granular flow around an object occurs in start-up/shut-down conditions in hopper/bin flows as well as during an impact event of an object into a granular soil. 3. Segregation due to granular shock waves will have a significant impact on the handling of materials in low gravity environments where dilute, granular shocks will be encountered.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	1. DEM computer simulations have been performed to show that the added mass of a cylinder in an accelerating dilute granular flow is negligible, a behavior that is considerably different from what is observed in ordinary liquid flows. A bow shock wave develops in these granular accelerating flows; however, the structure of this shock is insensitive to the flow speed and acceleration (again, a phenomenon much different from common compressible flows). A particularly interesting observation is that there is a significant difference in the unsteady and steady drag force during the period in which the bow shock wave develops. The bow shock is fully developed after a critical number of particle/cylinder impacts. Once the shock has formed, the unsteady and steady drag forces are identical. These results will have a significant impact on how the unsteady forces acting on immersed objects are treated in more complex flows. 2. Experiments concerning the low speed impact of spheres into horizontal granular beds have been performed and analytical models predicting the penetration depth have been developed. Previous work on this topic has generally been focused on developing empirical power law scaling relations for predicting the penetration depth of impacting objects. The work performed here, however, has focused on developing first-principles models for predicting the penetration depth. In particular, the penetration depth of a sphere normalized by the sphere's diameter is approximately proportional to the square root of the sphere density-to-granular bed bulk density ratio multiplied by the impact Froude number raised to the two-thirds power, where the impact Froude number is the impact velocity divided by the square root of the gravitational acceleration multiplied by the sphere's diameter. The analytical model accounts for the (static + dynamic) pressure and friction forces acting on the impacting sphere. Correlation between the model and experiments is excellent. Additional experiments have shown that the penetration depth is independent of the depth of the containing vessel as long as there is no contact between the sphere and the vessel. In contrast, the penetration depth decreases significantly when the containing vessel diameter approaches the diameter of the impacting sphere. 3. The dilute flow experiment (consisting of a two-dimensional, 4 x 8 ft air table with 1" plastic disks as particles), is being used to investigate dilute, two-dimensional granular flows. The air table design was developed in order to provide a low-cost, easily modifiable experiment for investigating flows where gravitational effects are negligible. Forces acting on a cylinder mounted on the air table are measured using a force transducer and using a momentum survey method. The flow is also videotaped for flow field analysis. Experimental data have been gathered using the apparatus with the measured results matching the simulations predictions at large Knudsen numbers. At smaller Knudsen numbers, however, the experiments and simulations diverge. The discrepancies have been determined to result from errors in the experimental data analysis. At small Knudsen numbers (corresponding to large solid fractions), the particle tracking algorithm used to determine the flow velocity field produces significant errors since incorrect tracks are frequently calculated. Furthermore, the collisional stress components, quantities that cannot be easily measured in the experiments, in the momentum wake survey analysis become significant. Measurements from the force transducer also are difficult to assess at larger solid fractions since individual impacts are no longer easy to differentiate from extraneous system oscillations.

Bibliography Type:	Description: (Last Updated: 04/18/2007)
Articles in Peer-reviewed Journals	Bharadwaj R, Wassgren C, Zenit R. "The unsteady drag force on a cylinder immersed in a dilute granular flow." Physics of Fluids 2006 Apr;18(4):043301. http://dx.doi.org/10.1063/1.2191907 , Apr-2006
Articles in Peer-reviewed Journals	Wassgren CR, Cordova JA, Zenit R, Karion A. "Dilute granular flow around an immersed cylinder. " Physics of Fluids 2003 Nov;15(11):3318-30. http://dx.doi.org/10.1063/1.1608937 , Nov-2003
Articles in Peer-reviewed Journals	Chehata D, Zenit R, Wassgren CR. "Dense granular flow around an immersed cylinder. " Physics of Fluids 2003 Jun;15(6):1622-31. http://dx.doi.org/10.1063/1.1571826 , Jun-2003
Presentation	Cordova, J.; Wassgren, C.; and Zenit, R. "Dilute granular flow around a cylinder" The 57th Annual Meeting of the Division of Fluid Dynamics, American Physical Society, Seattle, WA. Nov-2004
Presentation	Wassgren, C. "Granular flow around an immersed cylinder" The Gordon Conference on Granular and Granular-Fluid Flow, Colby College, ME. Jun-2004
Presentation	Wassgren, C. "Short course on Discrete Element Modeling" Universidad Nacional Autónoma de México Jun-2004

Presentation	Wassgren, C. "Modeling of Granular Flows Using the Discrete Element Method " Seminar at the Department of Industrial and Physical Pharmacy at Purdue University Feb-2004